STUDENT ACADEMIC PERFORMANCE REPORT AND COURSE SYLLABUS

Student Information

Particulars	Details
Student Name	Gautham Manoj
Roll Number	AM.ID.R4IDS23007-FT
Academic Program & Branch	Ph.D. 2023
Current CGPA	8.76

Semester Performance Details

Semester 4

Semester	Course Code	Course Name	Academic Term Period	Course Type	Grade Obtained
4	18MAT660	Graph Theory	2024–2025 Even Semester	Regular	A

Performance Summary

Metric	Score
SGPA	9.0
CGPA	8.76

Course Details

Course Code: 18MAT660

Course Title: Graph Theory

Credits: 3-0-0-3

Grade Obtained: A

COURSE OVERVIEW

Graph Theory is a graduate-level course offered by the Department of Mathematics, Amrita School of Physical Sciences, Amritapuri Campus, Amrita Vishwa Vidyapeetham. The course provides a mathematically rigorous treatment of graph structures with emphasis on connectivity, graph algorithms, matchings, colorings, and planar graphs. The concepts covered in this course have extensive applications in network analysis, optimization, operations research, computer science, artificial intelligence, transportation systems, and big data analytics.

COURSE SYLLABUS

UNIT I – FUNDAMENTALS OF GRAPHS

Graphs and Subgraphs, Isomorphism, matrices associated with graphs, degree of a graph, connected graphs, shortest path algorithm, trees, cut edges and cut vertices, spanning trees, and minimum spanning trees.

UNIT II – CONNECTIVITY AND EULERIAN/HAMILTONIAN GRAPHS

Graph connectivity, k-connected graphs and blocks, Euler graphs, Euler's Theorem, Fleury's Algorithm for Eulerian trails, Hamilton cycles, Chinese Postman Problem, and Traveling Salesman Problem.

UNIT III – MATCHINGS AND COVERINGS

Matchings, maximal matchings, coverings and minimal coverings, Berge's Theorem, Hall's Theorem, Tutte's Perfect Matching Theorem, Job Assignment Problem, Independent Sets, and Cliques.

UNIT IV – GRAPH COLORING

Vertex colorings, Greedy Algorithm and its consequences, Brooks' Theorem, edge colorings, and Vizing's Theorem on edge colorings.

UNIT V – PLANAR GRAPHS

Planar Graphs, Euler Formula, Dual Graphs, Kuratowski's Characterization of Planar Graphs, and Planarity Testing Algorithm.

TEXTBOOK

J. A. Bondy and U. S. R. Murty, *Graph Theory and Applications*, Springer.

REFERENCE BOOKS

1. **D. B. West**, *Introduction to Graph Theory*, PHI.
2. **Frank Harary**, *Graph Theory*, PHI.

COURSE RESULT SUMMARY

Course Code	Course Title	Semester	Credits	Grade	SGPA
18MAT660	Graph Theory	4	3-0-0-3	A	9.0

ACADEMIC RECORD

Student Name: Gautham Manoj

Roll Number: AM.ID.R4IDS23007-FT

Programme: Ph.D. 2023

Course Completed: Graph Theory (18MAT660)

Semester: 4

Grade Awarded: A

Semester GPA (SGPA): 9.0

Cumulative GPA (CGPA): 8.76

Academic Term: 2024–2025 Even Semester

Prepared as a consolidated academic record containing the student's performance details and complete syllabus for **18MAT660 – Graph Theory**.